

# Formal Review and Critique: Recursive Asymptotic Variance Estimation in CTMC

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## Executive Summary

The proposed algorithm for estimating the asymptotic variance  $\kappa(f)$  in a Continuous-Time Markov Chain (CTMC) setting is a logically sound extension of the discrete-time stochastic approximation (SA) framework. The pedagogical bridge provided in the derivation is commendable. However, for a formal MS thesis, the current draft exhibits several logical gaps, notation inconsistencies, and unaddressed stability concerns that must be rectified.

## 1 Scrutiny of Assumptions and Logical Gaps

### 1.1 State Space Dimensionality Conflict

The document fluctuates between treating the state space  $S$  as countable (Abstract) and finite (Algorithm 1 and Section 7).

- **Concern:** The use of  $1/S$  in the projection update  $V_{k+1}(x) = V_k(x) - \alpha_k \delta_k / S$  implicitly assumes that  $S$  is a finite integer representing  $|S|$ . In a countably infinite state space, this term is mathematically undefined.
- **Requirement:** Explicitly state the assumption  $|S| < \infty$  or generalize the projection operator  $\Pi$  using the stationary expectation  $\mathbb{E}_\pi[V]$  to handle the countable case.

### 1.2 Stochastic Step-Size Stability

The algorithm introduces a holding-time weighted update:  $\gamma_k = \alpha_k \tau_k$ , where  $\tau_k \sim \text{Exp}(q(X_k))$ .

- **Concern:** Standard SA theory (e.g., Borkar) relies on deterministic step-size sequences. Here,  $\gamma_k$  is a *random variable*. If the transition rates  $q(x)$  are small (i.e., long-tailed holding times), the variance of the updates may explode.
- **Challenge:** Provide a proof that the variance of the holding times  $\tau_k$  is uniformly bounded across  $S$  to satisfy the noise-suppression condition  $\sum \mathbb{E}[\gamma_k^2] < \infty$ .

### 1.3 Coupled SA System Constraints

The derivation treats  $c_1, c_2, c_3$  as arbitrary positive constants. In coupled linear systems (Agrawal et al., 2024), these constants are critical.

- **Concern:** The tracking error of the value function  $V_k \rightarrow V$  must be managed relative to the convergence of the stationary mean  $\bar{f}_k$ . You have not provided the ratio constraints or the eigenvalue analysis required to ensure the Hurwitz stability of the joint system.

## 2 Methodological and Mathematical Clarity

### 2.1 Initial Condition and Mixing Time Bias

The derivation of  $\kappa(f)$  in Section 4 assumes stationarity ( $X_0 \sim \pi$ ), yet the algorithm runs on a single trajectory starting from an arbitrary  $X_0$ .

- **Requirement:** Address the "burn-in" period or initial bias. A formal thesis must quantify how the chain's mixing time influences the finite-sample error of the  $\kappa_n$  estimate.

### 2.2 The "Invisible Diagonal" Assumption

The pedagogical argument for the disappearance of the negative term in CTMC is based on the Lebesgue measure of the diagonal being zero.

- **Scrutiny:** This holds only if the reward function  $f(X_t)$  is integrated over piecewise constant paths (càdlàg). If the underlying process included a diffusion component (e.g., reflected Brownian motion), the quadratic variation would not be zero on the diagonal. You must explicitly limit your scope to pure-jump CTMCs.

## 3 Notation and Formatting Inconsistencies

- **Standardization:** Standardize the use of boldface for vectors and matrices. Section 2 uses scalar notation ( $QV(x)$ ), whereas Section 7 correctly uses vector-matrix notation ( $A = \Pi D_\pi Q$ ). Consistency is mandatory for academic rigor.
- **Jump Chain Clarification:** In Section 6.2, explicitly define  $X_{k+1}$  as the state of the *embedded jump chain* to avoid confusion with the continuous-time evolution between transition points.

## Final Verdict

The algorithm is theoretically promising but mathematically underspecified. Addressing the stability of the **stochastic step-sizes** and providing **rigorous constraints for the constants**  $c_i$  are the highest priorities for the upcoming defense.